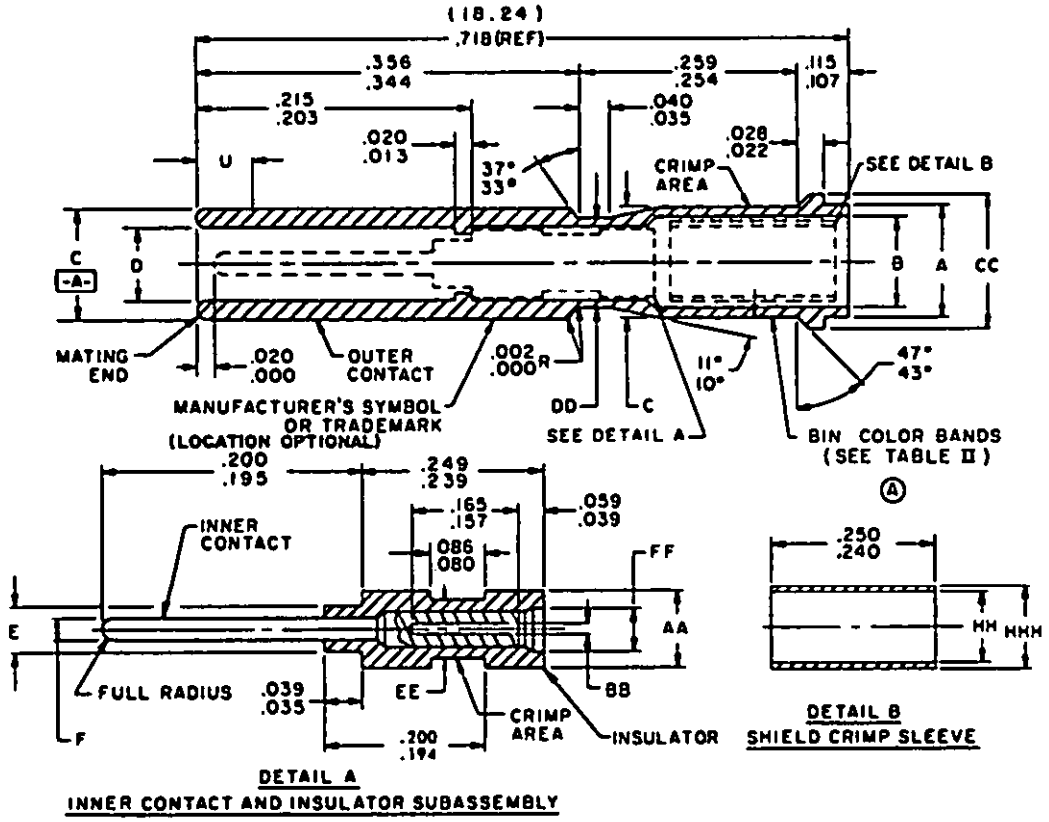
MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR,
SOCKET, CRIMP REMOVABLE, SHIELDED,
(FOR MIL-C-81511 SERIES 2 CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the contacts described herein shall consist of this document and the latest issue of Specification MIL-C-39029.

INACTIVE FOR NEW DESIGN



NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses for overall length and diameter only.
4. Dimensions shown apply after plating.

FIGURE 1. Connector contact.

FSC 5935

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TABLE I. Dimensions.

BIN code	A DIA	B DIA	C DIA	D DIA	E MAX DIA	AA DIA	BB MIN DIA	CC DIA	DD DIA	EE DIA	FF MIN DIA	HH MIN DIA	HHH MAX DIA	U
158	.103	.0910	.113	.0750	.042	.081	.022	.133 (3.38)	.099	.068	.037	.058	.087	.042
	.101	.0885	.110	.0725		.078		.130 (3.30)	.096	.062				.018
159	.162	.1430	.161	.1210	.070	.128	.022	.190 (4.83)	.145	.076	.067	.086	.130	.042
	.159	.1405	.158	.1175		.126		.187 (4.75)	.142	.070				.018
160	.162	.1430	.161	.1210	.070	.128	.022	.190 (4.83)	.145	.076	.108	.110	.130	.042
	.159	.1405	.158	.1175		.126		.187 (4.75)	.142	.070				.018
161	.162	.1430	.161	.1210	.070	.128	.022	.190 (4.83)	.145	.076	.108	.128	.142	.042
	.159	.1405	.158	.1175		.126		.187 (4.75)	.142	.070				.018
162	.162	.1430	.161	.1210	.070	.128	.034	.190 (4.83)	.145	.089	.098	.128	.142	.042
	.159	.1405	.158	.1175		.126		.187 (4.75)	.142	.085				.018
163	.162	.1430	.161	.1210	.070	.128	.034	.190 (4.83)	.145	.089	.098	.110	.130	.042
	.159	.1405	.158	.1175		.126		.187 (4.75)	.142	.085				.018

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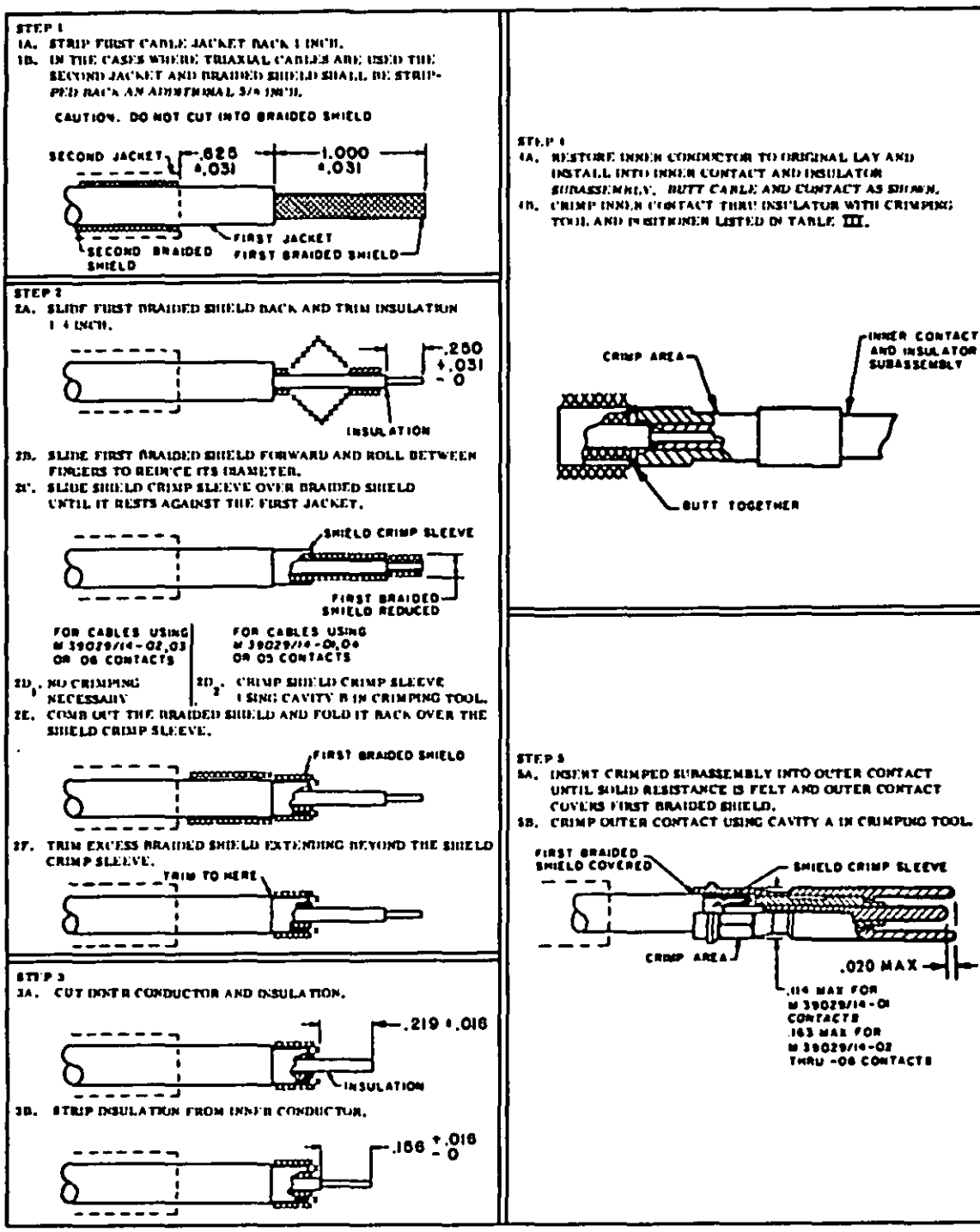


FIGURE 2. Assembly procedure.

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TABLE II. Design characteristics.

BIN code	Color bands			Contact cavity size	Cables accommodated	Type	Class
	1st	2nd	3rd				
158	Brown	Green	Gray	16	RG-178A/U RG-196A/U	D	B
159	Brown	Green	White	12	RG-179B/U RG-187A/U RG-188A/U		
160	Brown	Blue	Black	12	9530D5117 (RAYCHEM) 1/		
161	Brown	Blue	Brown	12	RG-180B/U RG-195A/U 293-3922 (MICRODOT) 1/		
162	Brown	Blue	Red	12	250-4070 (MICRODOT) 1/		
163	Brown	Blue	Orange	12	5022E5111 (RAYCHEM) 1/		

1/ or equivalent

REQUIREMENTS:

Dimensions, design characteristics, and configuration: See figure 1 and tables I and II.

Tools: See table III.

Mating contact: MIL-C-39029/6.

Manufacturer's recommended assembly instructions to be shipped with unit package.

Assembly procedure: See figure 2.

Preparation of samples: Contacts shall be wired as required using wire in accordance with table II (cables accommodated column).

Contact resistance: Contact resistance at a load current of 1.0 ampere shall meet the requirements of table IV.

Contact engagement and separation forces: The contact separation and engagement forces shall meet the requirements of table IV.

Dielectric withstanding voltage: 1,000 volts, ac, rms from sea level to 110,000 feet altitude.

Tensile strength: The tensile strength shall meet the requirements of table IV.

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TABLE III. Tools.

BIN code	Inner contact		Shield crimp sleeve		Outer contact		Installing tool	Removal tool
	Basic crimping tool	Positioner	Basic crimping tool	Die	Basic crimping tool	Die		
158	M22520/2-01	M22520/2-26	M22520/5-01	M22520/5-02 Cavity B	M22520/5-01	M22520/5-02 Cavity A	MS3323-16	MS3344-16
159	M22520/2-01	M22520/2-25	M22520/5-01	M22520/5-04 Cavity A	M22520/5-01 M22520/10-01	M22520/5-04 Cavity A M22520/10-04 Cavity A	MS3323-12	MS3344-12
160	M22520/2-01	M22520/2-25	M22520/5-01	M22520/5-04 Cavity A	M22520/5-01 M22520/10-01	M22520/5-04 Cavity A M22520/10-04 Cavity A	MS3323-12	MS3344-12
161	M22520/2-01	M22520/2-25	M22520/5-01 M22520/10-01	M22520/5-04 Cavity B M22520/10-04 Cavity B	M22520/5-01 M22520/10-01	M22520/5-04 Cavity A M22520/10-04 Cavity A	MS3323-12	MS3344-12
162	M22520/2-01	M22520/2-25	M22520/5-01 M22520/10-01	M22520/5-04 Cavity B M22520/10-04 Cavity B	M22520/5-01 M22520/10-01	M22520/5-04 Cavity A M22520/10-04 Cavity A	MS3323-12	MS3344-12
163	M22520/2-01	M22520/2-25	M22520/5-01	M22520/5-04 Cavity A	M22520/5-01 M22520/10-01	M22520/5-04 Cavity A M22520/10-04 Cavity A	MS3323-12	MS3344-12

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TABLE IV. Test limits.

BIN code	Tensile strength (lbs) min		Contact engaging and separation forces (ounces)				Maximum contact resistance (millivolts)	
			Inner contact		Contact assy			
	Inner contact	Outer contact	Min	Max	Min	Max	Inner contact	Outer contact
158	6.0	15	0.5	7.0	2.0	20.0	8.0	4.0
159	6.0							2.0
160	16.0							
161	6.0							
162	32.0							
163	12.0							

QPL evaluating activity: Naval Weapon Support Center, Code 3074, Crane, Indiana, 47522.

Military part numbers: See table V.

TABLE V. Military part number.

Bin code	Military part number
158	M39029/14-01
159	M39029/14-02
160	M39029/14-03
161	M39029/14-04
162	M39029/14-05
163	M39029/14-06

Custodians:
Navy - AS

Review activities:
Navy - EC, SH, MC
DSA - ES

User activities:
Navy - OS

Preparing activity:
Navy - AS

Agent:
DSA - ES

(Project 5935-N041-2)

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